Verifiable Private Governance on Ethereum

Closing the Acceptance Gap for Private Collective Decisions

Ethereum Foundation PhD Fellowship 2026 — Proposal
RFP: Programmable Institutional Design and Verifiable Governance

Vincenzo Imperati
PhD Student in Computer Science, Sapienza University of Rome
imperati@di.uniroma1.it · vincenzo.imperati.dev

Project Abstract

Ethereum can encode governance rules precisely, but private collective decisions still often rely on trusted coordinators, tally operators, or workflow controls that are hard to audit. This project studies a reusable mechanism for closing the acceptance gap between an encrypted decision and the proof that authorizes it, even when the encryption and proof systems are cryptographically incompatible. The flagship case is private voting and committee decision-making using threshold-Paillier for efficient encrypted aggregation and Groth16 for proof verification that Ethereum can perform cheaply on-chain. The fellowship would fund a stronger aggregate-integrity mechanism built on top of that acceptance-layer base, a governance-level model of what is immutable, adjustable, and discretionary, and a public artifact plus open-access paper for Ethereum governance builders.

Objectives

1. **Formalize a reusable mechanism for private collective decisions.** Define clearly which institutional constraints become feasible through programmability and verifiability in this setting, and map the design into immutable, adjustable, and discretionary components, including who has authority to set or change each part of the workflow.
2. **Strengthen tally-side integrity in a realistic way.** Starting from the current transcript-auditable construction, design, implement, and formally analyze an aggregate-integrity mechanism that reduces reliance on privileged tally actors and closes the most important gap between accepted encrypted ballots and the published aggregate.
3. **Produce reusable public infrastructure.** Release a public artifact for Ethereum builders, including the core circuit and contract logic, reproducible benchmarks, documentation, and a small reusable developer surface.
4. **Publish an open-access academic output.** Prepare and submit the research to a named security venue target, with a public preprint path, archived artifact, and permissive open-source licensing.

Success metrics.

- One paper submitted during the fellowship period, with a documented submission record and archived public artifact
- One maintained public artifact release with documentation and reproducible results

- One implemented and evaluated aggregate-integrity mechanism stronger than the current transcript-auditable baseline
- One external round of feedback from Ethereum governance practitioners or adjacent projects

Outcomes

From trusted procedure to enforceable rule

The practical contribution is not "blockchain voting" in the abstract. It is a reusable mechanism for higher-assurance private collective decisions on Ethereum: committee votes, council approvals, board or membership decisions, sealed governance proposals, and similar workflows where participants need confidentiality without giving up public verifiability. The project targets institutional and organizational settings, not national political elections [8].

In a traditional confidential decision process, several roles are usually separated: an administrator manages eligibility, trusted staff or scrutineers handle ballot custody, and tally trustees or operators ensure that the object counted is the same one originally cast. In many current Ethereum privacy architectures, the same acceptance-side guarantee still falls to a coordinator or application-specific off-chain procedure. The main institutional outcome of this project is to move one specific rule from privileged trust into verifiable code: once an encrypted decision is accepted, the object recorded on-chain must be the same object authorized by the proof.

A clearer allocation of authority

This project speaks directly to the RFP's emphasis on predictable, enforceable, and selectively controllable governance structures. In the governance design studied here:

- **Immutable:** the acceptance relation between the encrypted object and the proof that authorizes it; no coordinator, administrator, or tally actor should be able to override that relation after acceptance
- **Adjustable:** governance parameters such as eligibility roots, thresholds, election timing, tally rules, and pause or restart policy, which remain under institutional control before or between governance rounds
- **Discretionary:** optional modules above the cryptographic core, such as ballot-validity policies, decryption ceremonies, operational review, or institution-specific exception handling

This is intentionally a hybrid design rather than a fully automated one. Programmability is used where discretionary override is most dangerous, while institutions retain human control where policy, process, or legal context may legitimately require it. The result is not only privacy, but a more legible allocation of control: auditors and governance designers can see which rules are hard-coded, which can be updated by authorized governance actors, and which are left to explicit operational discretion.

Real governance and compliance problems addressed

The immediate problems addressed are not theoretical only. Many organizations need procedures in which individual decisions remain confidential until a process closes, while outsiders can still verify that only eligible inputs were accepted and that the published aggregate corresponds to those accepted inputs. This arises in grant committees, professional associations, protocol councils, university bodies, and other collectives that need confidentiality without opaque administration.

The project therefore treats auditable governance and verifiable compliance as core features rather than side benefits.

Benefit to the wider Ethereum ecosystem

Ethereum has strong governance tooling, but privacy-preserving governance still often relies on specialized workflow actors, off-chain coordination, or design tradeoffs that are difficult to reuse. This project benefits the ecosystem in three ways:

- it turns a currently under-specified mixed-domain composition problem into a documented and reusable design pattern;
- it gives Ethereum builders a public artifact and benchmark baseline for a governance problem that sits directly at the intersection of encryption, succinct proofs, and EVM constraints shaped by Ethereum's BN254 precompiles [1, 2];
- it translates a narrow cryptographic contribution into institutional design language that can be understood by non-cryptographers working on governance products and frameworks.

Grant Scope

Current starting point

This proposal does not start from a blank slate. I already have a public research artifact centered on the acceptance side of the problem, including the acceptance gap between encrypted decisions and their on-chain authorization proof: a Circom circuit, a Solidity contract using a real Groth16 verifier on EVM, a reproduced attack on the naive composition, passing tests, and measured benchmarks at realistic parameters. I also have a paper draft and a detailed map of claims, evidence, and open gaps.

This existing base reduces execution risk, but the work is not already complete. The current artifact is strongest on acceptance-side integrity and on public auditability of the tally transcript. The fellowship would fund the next substantive step: turning that base into a clearer governance contribution with stronger tally-side guarantees, public packaging, and a publication-ready research output.

Research work funded by the fellowship

Work Package 1: Formal model and governance framing. I will refine the current technical model into a governance-oriented formal description of the mechanism, including:

- the mixed-domain threat model;
- the mapping between cryptographic guarantees and institutional constraints;
- a clear immutable / adjustable / discretionary decomposition with a specified authority model; and
- a comparison with both traditional trustee- or coordinator-run workflows and nearby Ethereum governance architectures.

Work Package 2: Aggregate-integrity mechanism. I will design, implement, and evaluate a mechanism that strengthens the current tally path beyond mismatch detectability. The primary

path is an incremental aggregate-integrity construction in which the tally-side aggregate becomes a deterministic function of accepted ciphertexts by construction, reducing room for operator discretion after acceptance. If that primary path proves too costly or too weak for a precise claim, the fallback is a narrower tally-consistency upgrade that still materially improves verifiability over the present baseline.

Any retained construction must satisfy four conditions:

- it yields a precise formal claim;
- it has clear implementation semantics;
- it can be tested and benchmarked convincingly;
- it does not distort the project away from its governance-mechanism core.

Work Package 3: Public artifact and ecosystem packaging. I will mature the existing public repository into a reusable artifact release whose packaging is centered on the artifact itself rather than on the broader research history behind it. This will include:

- the core circuit and contract logic;
- benchmark data and regeneration instructions;
- threat-model and limitation notes;
- developer-facing documentation; and
- a small reusable package or integration surface for Ethereum builders.

Work Package 4: Academic output and dissemination. I will write and submit the paper, target IEEE S&P 2027 Cycle 2 as the primary venue and ACM CCS 2027 as fallback, archive the artifact, and produce a concise non-technical explanation for Ethereum governance practitioners.

Expected outputs

- One open-access research paper submitted to IEEE S&P 2027 Cycle 2 or ACM CCS 2027
- One public preprint or equivalent open-access manuscript path
- One documented public open-source artifact release under a permissive license
- One archived artifact snapshot, for example Zenodo
- One concise practitioner-facing technical explainer for the Ethereum governance ecosystem

Related Work

This project sits between institutional governance practice and cryptographic systems work that rarely meet cleanly in Ethereum deployments.

On the governance side, confidential decision workflows traditionally rely on a separation of duties among administrators, scrutineers, or tally trustees. Ethereum already has mature systems for collective decision-making, ranging from off-chain signaling to on-chain execution frameworks, but private governance architectures encode or retain those roles in different ways. MACI shows that private on-chain governance is valuable and feasible, but its audited architecture still organizes tallying and proof generation around a coordinator role. Open Vote Network shows a different design point through self-tallying voting, avoiding the threshold-encrypted aggregation path altogether.

Helios and the follow-up critique literature remain important reminders that composition-level gaps, not only broken primitives, can undermine governance systems [3, 4, 5, 6, 7].

On the cryptographic side, same-domain composition is much better understood. LegoSNARK provides a powerful modular composition framework inside a proof-friendly domain. SAVER and newer encryption-friendly proof systems such as Tangram illustrate how much easier composition becomes when encrypted objects and proof statements remain within compatible algebraic structures. Recent work on Paillier proofs, including efficient zero-knowledge arguments and direct range proofs, shows that Paillier-side reasoning is possible, but also that it is not free in practice. Modern threshold Paillier systems such as Tiresias show that Paillier remains a serious primitive rather than only a historical one [9, 10, 11, 12, 13, 14].

This broader relevance is not unique to voting. In federated learning, secure aggregation, verifiable aggregation, and input verification are active problems, which reinforces the point that encrypted aggregation plus public or third-party-checkable correctness is an important systems pattern in its own right [15, 16]. I am not proposing a full federated learning deliverable in this fellowship, but the literature helps show that the composition pattern isolated here is not an eccentric one-off artifact of voting.

The specific gap addressed here is narrower and more practical: Ethereum still lacks a clean, reusable mechanism for the case where a raw encrypted governance object must be authorized by a proof that Ethereum can verify cheaply on-chain, despite the encryption and proof systems living in incompatible cryptographic domains. That is the acceptance gap I aim to formalize and close. Existing systems either remain in same-domain cryptographic settings, avoid threshold-encrypted aggregation altogether, or accept different workflow assumptions. The project therefore does not claim that encryption, succinct proofs, or private governance are individually missing; it claims that their mixed-domain interface on Ethereum remains under-specified and under-packaged.

About the Applicant

I am a PhD student in Computer Science at Sapienza University of Rome, where I work under the supervision of Alessandro Mei. My broader PhD work is in security, large-scale systems, and open protocols, and I am currently a Visiting Researcher at the University of Zurich. I have published at *USENIX Security* and *IEEE Access*, and I have spent the last years building public infrastructure as well as doing academic research. This proposal sits inside my PhD research environment rather than outside it: the fellowship would fund a line of work already grounded in university research, existing supervision, and a working public artifact.

I am well suited to this fellowship because this project requires exactly the combination I already have: security-research discipline, real artifact engineering, and practical Ethereum experience. I founded and maintain *BigBrotr*, an OpenSats-funded observability platform for the Nostr protocol, which demonstrates my ability to deliver funded public research infrastructure. I have also built multiple EVM projects, including MeltyFi, Nostreum, and the earlier on-chain election-integrity prototype from which this research direction emerged. This means I am not approaching Ethereum governance as an external observer; I am already an active builder in the ecosystem.

Relevant links

- Personal website: vincenzo.imperati.dev
- Sapienza researcher profile: research.uniroma1.it

- GitHub: github.com/VincenzoImp
- USENIX Security paper: The Conspiracy Money Machine
- IEEE Access paper: Hard to See, Harder to Block
- Project repository link: github.com/VincenzoImp/paillier-groth16-binding
- ORCID: 0009-0001-9437-1384
- Google Scholar: scholar.google.com/citations?user=V4XgGVcAAAAJ
- Supervisor academic profile: Alessandro Mei

Methodology

This is applied cryptography research with a concrete institutional example. The methodological goal is not to simulate a full voting platform; it is to study, strengthen, and package one reusable governance mechanism that becomes useful once private decisions and succinct authorization proofs must coexist on Ethereum.

Phase 1: Formalization and architecture comparison

I will start by refining the mixed-domain model from the current research base into a cleaner governance-level statement. This includes:

- specifying the threat model induced by a public bulletin board and cheap on-chain proof verification;
- mapping the mechanism into immutable, adjustable, and discretionary governance components together with the authority structure around each;
- comparing it against adjacent design points, especially coordinator-centric, self-tallying, and same-domain approaches; and
- making explicit the comparison with traditional trustee- or scrutineer-based governance workflows.

Phase 2: Design and evaluation of the aggregate-integrity upgrade

I will then build the main research extension of the fellowship: a stronger aggregate-integrity mechanism. I already know what the present baseline is:

- accepted ballots are bound correctly at the board layer;
- transcript-derived aggregate mismatch is publicly detectable.

The fellowship will fund the step beyond that baseline. I will first implement the primary path described in the Grant Scope: an incremental aggregate-integrity mechanism that makes the published aggregate follow from the accepted ciphertext set more directly. If that path does not support a clear and credible claim after implementation and measurement, I will fall back to a narrower tally-consistency upgrade rather than forcing a weak result into the paper.

Phase 3: Public artifact, documentation, and paper

Once the research line is stable, I will:

- stabilize the existing public repository;
- write a clear threat-model and limitations document;

- archive benchmark outputs and regeneration paths;
- prepare the artifact for archival release; and
- produce a short explainer for practitioners who care about governance design but are not specialists in cryptography.

Phase 4: Submission and dissemination

In the final phase, I will turn the research into a formal academic output and a usable ecosystem deliverable. This includes rewriting the paper into venue-ready form, submitting it to the target venue, preserving a public manuscript path consistent with venue policy, and collecting external feedback from adjacent Ethereum governance builders.

Hybrid design and scope discipline

The project is intentionally hybrid rather than fully automated. Acceptance-side binding and the corresponding aggregate-integrity mechanism are the parts that should become maximally rigid and auditable. Decryption ceremonies, institution-specific ballot policies, and exceptional operational decisions may remain partially off-chain or discretionary, because the purpose of the project is to clarify the boundary between programmable enforcement and retained human control, not to eliminate institutions from the loop.

The project will remain intentionally scoped. During the fellowship I will not reframe it as:

- a complete national-election system;
- a complete solution to all tally and decryption correctness questions;
- or a three-application general framework spanning voting, federated learning, and MPC simultaneously.

This scope discipline is part of the execution strategy: the goal is to deliver one strong, useful result rather than many half-finished ones.

Timeline

Planned fellowship period:

- June 2026 to May 2027
- total requested fellowship support: $24,000
- approximate total effort used for planning: 720 hours
- indicative planning rate used for milestone allocation: approximately $33.33/hour
- milestone budgets below distribute the fixed fellowship amount across the planned work packages

Milestone 1: Institutional model and architecture comparison

Timing: Months 1–3
Budget: $6,000
Number of hours (roughly): 180

Work

- consolidate the formal and governance model
- finalize the immutable / adjustable / discretionary decomposition
- compare the mechanism against trustee-run, coordinator-run, and adjacent Ethereum governance designs

Deliverables

- stable problem statement
- governance-comparison section including traditional workflow counterpart
- updated research plan for the aggregate-integrity extension

Milestone 2: Aggregate-integrity mechanism

Timing: Months 3–6
Budget: $8,000
Number of hours (roughly): 240

Work

- implement the primary aggregate-integrity path
- implement the mechanism in the existing artifact stack
- add tests and measure cost
- formalize the corresponding claim, or narrow the result to the strongest defensible fallback

Deliverables

- implemented and benchmarked aggregate-integrity upgrade
- updated threat model and limitation notes
- locked final paper scope

Milestone 3: Public artifact stabilization and archival release

Timing: Months 6–8
Budget: $5,000
Number of hours (roughly): 150

Work

- stabilize the existing public repository
- write documentation and reproduction instructions
- package benchmark data and artifact metadata
- archive a stable tagged snapshot

Deliverables

- documented public artifact release
- archived release snapshot
- practitioner-facing explainer draft

Milestone 4: Paper submission and dissemination

Timing: Months 8–12
Budget: $5,000
Number of hours (roughly): 150

Work

- complete the paper
- submit to IEEE S&P 2027 Cycle 2 or ACM CCS 2027 fallback
- collect ecosystem feedback
- publish the public explainer and integration notes

Deliverables

- submitted paper and submission record
- public preprint path or submission-ready manuscript
- final public explainer
- short feedback memo from ecosystem conversations

References

[1] Christian Reitwiessner. EIP-196: Precompiled contracts for addition and scalar multiplication on the elliptic curve alt_bn128. Ethereum Improvement Proposals, no. 196, 2017. https://eips.ethereum.org/EIPS/eip-196

[2] Vitalik Buterin and Christian Reitwiessner. EIP-197: Precompiled contracts for optimal ate pairing check on the elliptic curve alt_bn128. Ethereum Improvement Proposals, no. 197, 2017. https://eips.ethereum.org/EIPS/eip-197

[3] Kyle Charbonnet and Yufei Li. MACI Security Audit. PSE Security, July 2024. https://maci.pse.dev/assets/files/20240731_PSE_Audit_audit_report-a0a0f08a5c621ccd0389d5e345c119be.pdf

[4] Patrick McCorry, Siamak F. Shahandashti, and Feng Hao. A smart contract for boardroom voting with maximum voter privacy. In *Financial Cryptography and Data Security*, pages 357–375, 2017. doi: https://doi.org/10.1007/978-3-319-70972-7_20

[5] Ben Adida. Helios: Web-based Open-Audit Voting. In *17th USENIX Security Symposium*, pages 335–348, 2008. https://www.usenix.org/conference/17th-usenix-security-symposium/helios-web-based-open-audit-voting

[6] Véronique Cortier and Ben Smyth. Attacking and fixing Helios: An analysis of ballot secrecy. *Journal of Computer Security*, 21(1):89–148, 2013.

[7] David Bernhard, Véronique Cortier, David Galindo, Olivier Pereira, and Bogdan Warinschi. SoK: A comprehensive analysis of game-based ballot privacy definitions. In *IEEE Symposium on Security and Privacy*, pages 499–516, 2015. doi: https://doi.org/10.1109/SP.2015.37

[8] Sunoo Park, Michael Specter, Neha Narula, and Ronald L. Rivest. Going from bad to worse: from Internet voting to blockchain voting. *Journal of Cybersecurity*, 7(1):tyaa025, 2021. doi: https://doi.org/10.1093/cybsec/tyaa025

[9] Matteo Campanelli, Dario Fiore, and Anaïs Querol. LegoSNARK: Modular design and composition of succinct zero-knowledge proofs. In *ACM CCS*, pages 2075–2092, 2019. doi: https://doi.org/10.1145/3319535.3339820

[10] Jiwon Lee, Jaekyoung Choi, Jihye Kim, and Hyunok Oh. SAVER: SNARK-friendly, additively homomorphic, and verifiable encryption and decryption with rerandomization. Cryptology ePrint Archive, Paper 2019/1270.

[11] Gweonho Jeong, Myeongkyun Moon, Geonho Yoon, Hyunok Oh, and Jihye Kim. Tangram: Encryption-friendly SNARK framework under Pedersen committed engines. Cryptology ePrint Archive, Paper 2025/540.

[12] Borui Gong, Wang Fat Lau, Man Ho Au, Rupeng Yang, Haiyang Xue, and Lichun Li. Efficient Zero-Knowledge Arguments for Paillier Cryptosystem. Cryptology ePrint Archive, Paper 2024/1303.

[13] Zhikang Xie, Mengling Liu, Haiyang Xue, Man Ho Au, Robert H. Deng, and Siu-Ming Yiu. Direct Range Proofs for Paillier Cryptosystem and Their Applications. Cryptology ePrint Archive, Paper 2024/1355.

[14] Offir Friedman, Avichai Marmor, Dolev Mutzari, Yehonatan C. Scaly, Yuval Spiizer, and Avishay Yanai. Tiresias: Large Scale, Maliciously Secure Threshold Paillier. Cryptology ePrint Archive, Paper 2023/998.

[15] Xiaojie Guo. Fixing Issues and Achieving Maliciously Secure Verifiable Aggregation in "VeriFL: Communication-Efficient and Fast Verifiable Aggregation for Federated Learning". Cryptology ePrint Archive, Paper 2022/1073.

[16] Zhi Lu and Songfeng Lu. LZKSA: Lattice-based special zero-knowledge proofs for secure aggregation's input verification. Cryptology ePrint Archive, Paper 2025/1141.